Ankan Deria

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LinkedIn GitHub Portfolio Google Scholar

Research Interests

Researcher in multimodal AI working on **world models and long-horizon planning**, video generation, vision-language models, and unified multimodal architectures. **11 papers** (7 accepted, 4 under review; **8 first-author**) at venues including NeurIPS, CVPR, ACM-MM and AAAI. Currently an M.Sc. student in Computer Vision at MBZUAI, advised by Prof. **Salman Khan**.

Education

Mohamed bin Zayed University of Artificial Intelligence — **Aug 2025 – Expected 2027**
*M.Sc. in Computer Vision; Advisor: Prof. **Salman Khan**; GPA: 3.7/4 (First Year)* — *Abu Dhabi, UAE*

Government College of Engineering and Leather Technology — **Aug 2020 – Jul 2024**
B.Tech. in Computer Science and Engineering; CGPA: 9.26/10 — *Kolkata, India*

Publications

WorldGuide: Goal-Directed Video World Model for Procedural Task Execution
Authors: **Ankan Deria**, K. Kumar, H. Cholakkal, F. S. Khan, S. Khan
Under Review — ICLR 2027 ·

Training-Free Speech-Centric Omni Understanding with Frozen VLMs
Authors: **Ankan Deria**, H. Rasheed, X. He, F. S. Khan, S. Khan
Under Review — AAAI 2027 ·

SafeDiffusion-R1: Online Reward Steering for Safe Diffusion Post-Training
Authors: K. Kumar, **Ankan Deria**, A. Basu, F. Shamshad, H. Cholakkal, K. Nandakumar
Under Review — AAAI 2027 ·

OmniSE: Support-Faithful Optimization for Evidence-Centric Audio-Video Reasoning
Authors: X. He, Y. Shen, **Ankan Deria**, X. Yue, M. H. Khan
Accepted — NeurIPS 2026

CoME-VL: Scaling Complementary Multi-Encoder Vision-Language Learning
Authors: **Ankan Deria**, K. Kumar, X. He, I. Razzak, H. Cholakkal, F. S. Khan, S. Khan
Accepted — ACM-MM 2026 ·

MedMO: Grounding and Understanding Multimodal Large Language Model for Medical Images
Authors: **Ankan Deria**, K. Kumar, A. M. Dukre, E. Segal, S. Khan, I. Razzak
Accepted — CVPR 2026 ·

NeMoColor: Unified Noise Modulation for Enhanced Diffusion-based Image Colorization
Authors: **Ankan Deria**, D. Mahapatra, M. Mondal, S. Roy
Accepted — AAAI 2026 ·

ViMaR: Dual-Stage Value-Guided Inference with Margin-Based Reward Adjustment for Fast and Faithful VLM Captioning
Authors: **Ankan Deria**, A. M. Dukre, F. Tang, S. Atito, S. Roy, M. Awais, M. H. Khan, I. Razzak
Accepted — NeurIPS 2025 ·

MuGa-VTON: Multi-Garment Virtual Try-On via Diffusion Transformers with Prompt Customization
Authors: **Ankan Deria**, D. Mahapatra, B. Bozorgtabar, M. Chakraborty, S. Chakraborty, S. Roy
Under Review — TMLR 2026 ·

Robust semantic learning for precise medical image segmentation
Authors: S. Chakraborty, K. Kumar, **Ankan Deria**, D. Mahapatra, B. Bozorgtabar, S. Roy
Accepted — Biomedical Signal Processing and Control, 2025 ·

InVERGe: Intelligent Visual Encoder for Bridging Modalities in Report Generation
Authors: **Ankan Deria**, K. Kumar, S. Chakraborty, D. Mahapatra, S. Roy
Accepted — CVPRW 2024 ·

Research Experience

Apertix AI **May 2026– Jun 2026**
Research Intern *Abu Dhabi, UAE*

- Built an edge-deployable omni (speech, audio, vision) model from **Training-Free Speech-Centric Omni Understanding with Frozen VLMs** (under review, AAAI 2027).
- Developed an **iPhone app** running the omni model on-device, taking the research to a working edge demo.

Mohamed bin Zayed University of Artificial Intelligence **Aug 2025– Present**
M.Sc. Student, advised by Prof. ***Salman Khan*** *Abu Dhabi, UAE*

- **WorldGuide:** goal-directed video world model for long-horizon procedural tasks (under review, ICLR 2027).
- **MedMO:** multimodal LLM for medical image grounding and understanding (CVPR 2026; Compute Transparency Champion).
- **CoME-VL:** complementary multi-encoder vision-language learning (ACM-MM 2026).

Mohamed bin Zayed University of Artificial Intelligence **Apr 2025– Aug 2025**
Research Assistant, advised by Prof. ***Imran Razzak*** *Abu Dhabi, UAE*

- **ViMaR:** value-guided inference with margin-based rewards for fast, faithful VLM captioning (NeurIPS 2025).

Jio Institute **Jul 2024– Apr 2025**
Researcher, advised by Prof. ***Sudipta Roy*** *Navi Mumbai, India*

- **NeMoColor:** noise modulation for diffusion-based image colorization (AAAI 2026).
- **MuGa-VTON:** diffusion-transformer multi-garment virtual try-on (under review, TMLR); medical image segmentation (BSPC 2025).

Jio Institute **Jul 2023– Oct 2023**
Research Intern, advised by Prof. ***Sudipta Roy*** *Navi Mumbai, India*

- **InVERGe:** visual encoder for radiology report generation (CVPRW 2024).

Leadership

Omdena **Dec 2022– Aug 2023**
Chapter Lead *Remote*

- Led the Kolkata chapter's open-source NLP projects: **Text Summarization**, **Hate Speech Detection**, and **Job Recommendation System**.

Honors and Awards

- **CVPR 2026 Compute Transparency Champion Award** (for MedMO)
- **3rd place**, MBZUAI Research Showcase 2025
- **69th rank**, Amazon ML Challenge, 2023; selected for Amazon ML Summer School, 2022
- **Reviewer:** NeurIPS, ICLR, WACV, BMVC

References

Prof. Salman Khan
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Prof. Fahad Khan
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Dept. of Computer Vision
MBZUAI, Abu Dhabi, UAE
Email: fahad.khan@mbzuai.ac.ae

Prof. Sudipta Roy
Associate Professor
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Jio Institute, Navi Mumbai, India
Email: sudipta1.roy@jioinstitute.edu.in